VORA: streaming voice Q&A on CPU (ASR, RAG, LLM, TTS)

Free, open-source models, CPU only, English and Mandarin. Numbers come from `results/*.json` (final run 2026-10-06, Apple M2, 8 cores, test splits read once with `--final`), except design-choice figures that cite `docs/spikes.md`. Full design notes and every earlier table: `docs/appendix.md`.

1. Architecture

The browser microphone sends 16 kHz PCM16 audio in 100 ms frames over a WebSocket to the streaming ASR. Partial text goes to the UI, and stable partials start retrieval early. Retrieval is hybrid (vectors plus BM25, with a calibrated refusal threshold). The LLM streams tokens into an answer guard (negation, figures, refusals answered from the retrieved text). A chunker cuts the guarded text into 1 word, then 3 words, then sentences, and the TTS sends 16 kHz audio back on the same socket.

Each user gets one socket; ASR, retrieval, LLM and TTS run on separate executors with a bounded output queue, so a slow client only blocks its own TTS. Speaking over the answer cancels it (barge-in).

2. Gates (brief targets)

Gate	Target	Result	Measured
G1 latency, synthetic en audio	p50 ≤1500, p90 ≤1800 ms	PASS	p50 1006 / p90 1438 ms
G1r latency, real voices (MInDS-14)	answered p50 ≤1500, p90 ≤1800 ms	PASS	p90: en-US 1337, en-GB 1422, en-AU 1725, zh 1507 ms (§3)
G2 TTS first chunk	p95 ≤200 ms	PASS	en 70 / zh 89 ms
G3 ASR+RAG+TTS memory	≤500 MB USS	PASS	305 MB (macOS); 465 MB in the Linux container
G4 ASR accuracy, clean read speech	WER/CER ≤15%	PASS	en LibriSpeech 8.4%, zh AISHELL 5.8%
G5 retrieval + faithfulness	top-3 ≥80%, faithful ≥95%	FAIL	top-3 96%, faithful 89% of 100 (blind 81.7%)
G6 two users at once	each p50 ≤2× single	PASS	×1.36 (1034 / 1215 vs 891 ms)
G7 Docker + Pi-class proof	arm64, amd64, limited cores, Pi-class CPU	UNVERIFIED	3 of 4 done; Pi-class run refused by AWS free plan
G8 permissive licences	no unknown licences	FAIL	zh voice huayan: licence unknown (kept by owner ruling)

Latency = speech end to first audio of the answer, through the real models. LLM memory is separate: 653 MB.

3. Real human speech (third-party, not self-authored)

Questions and audio come from MInDS-14 (PolyAI, CC-BY-4.0): real callers asking a bank about 14 intents, recorded over the phone at 8 kHz. They are answered from a separate demo bank knowledge base written from the intent names before any utterance was read. Accuracy is on 100 test clips per cell; latency on 42 fixed clips per language (14 per English accent).

Scene	ASR error	Top-3 (ASR text / exact text)	Faithful (ASR / exact text)	Answered p50 / p90 ms (n)
en-US phone, 8 kHz	WER 44.0%	90% / 98%	52% / 67%	1061 / 1337 (13 answered, 1 refused)
en-GB phone	WER 49.1%	74% / 93%	42% / 57%	1178 / 1422 (10, 3)
en-AU phone	WER 42.0%	76% / 96%	43% / 56%	1034 / 1725 (11, 3)
zh-CN phone	CER 21.5%	83% / 92%	59% / 60%	1006 / 1507 (28, 14)

Latency counts answered turns, as the gate defines it; refused turns get a fixed short reply and are 7 to 33% of the latency clips.

Phone speech is hard for 16 kHz models at the 50 MB size limit, yet retrieval still finds the right answer most of the time because a few key words are enough. Babble and reverb are far worse (en-US WER 77.5% and 92.3%, top-3

70% and 8%; full noise table in the appendix), and under reverberation the system refuses (77% refused) instead of guessing.

4. Latency stages, RTF and MOS

English audio, final run (26 turns, quiet host, the same turns as G1):

Stage	p50 ms	p90 ms
ASR (speech end to final text)	910	1123
RAG + LLM first token (after final text)	0	147
TTS first chunk (after first token)	316	676
Total (speech end to first audio)	1006	1438

The rows overlap, so they do not add up: a shadow turn starts the answer on a stable partial and hides most retrieval and LLM time inside the endpoint wait (faster on 22 of 25 questions in a paired test, median gain 153 ms). The LLM alone needs 438 ms to its first token; the cost without the shadow turn was not measured.

Real-time factor (processing time over audio length) on the M2: ASR English 0.053 to 0.076 and Chinese 0.028 to 0.034, at least 13 times faster than real time. English TTS has a clause RTF of 0.08 (`docs/spikes.md`); Chinese TTS RTF was not recorded, and nothing was measured on a Raspberry Pi 4.

MOS proxy (UTMOS22, 10 clips per voice): English 4.22, Chinese 3.44, against the 3.5 target. UTMOS is an English-trained predictor, not a listening test.

5. Stack choices

Part	Choice	Why
ASR	sherpa-onnx zipformer en-20M int8 (41.6 MB), small CTC zh int8 (25 MB)	Both fit 50 MB. The CTC model beat the 14M transducer on the same 60 AISHELL-1 clips (CER 5.2% vs 16.7%); the bilingual model's 182 MB encoder is too big.
Retrieval	per-language int8 bge-small, FAISS plus BM25	BM25 keeps codes such as VORA-X200 retrievable. Int8 embedders and a numpy resampler cut Linux memory from 560 to 465 MB.
LLM	Qwen2.5-0.5B Q4_K_M via llama.cpp (469 MB)	Under 1B, Apache-2.0. GGUF has the lowest first token (438 ms vs 628 ms for ONNX int8). ONNX int8 decodes faster but is 603 MB, so it was benchmarked, not shipped.
TTS	Piper VITS: en LJSpeech (22.3 MB, int8 with fp32 decoder), zh huayan (20 MB)	Kitten was 3x slower; fp32 Piper is 63 MB; all-int8 LJSpeech had a 145 / 179 ms first chunk against 57 / 79 ms with the fp32 decoder.
Transport	FastAPI WebSocket, one socket per user	One connection carries audio both ways, with per-user backpressure; simpler than gRPC or WebRTC.

6. Trade-offs

- Model size against accuracy: the 50 MB ASR limit gives models that are strong on read speech (WER 8.4% en, CER 5.8% zh) and weak on phone speech.
- Latency against naturalness: the first audio chunk is one word, then three, then sentences. That keeps English first audio at p95 70 ms and costs prosody at the start.
- Endpoint delay against cut-off sentences: 0.4 s of trailing silence gives about half the extra finals of 0.3 s.
- Refusal threshold against recall: a higher threshold refuses more off-topic questions and also some real ones with ASR errors.
- Speculation against CPU: the shadow turn needs 6 or more cores and one session, so a 4-core Pi runs without it.

- Gain control against clean-speech accuracy: it fixes quiet speakers (WER 42.9% to 5.6% at −30 dB) and costs at most 1.3 points elsewhere.

7. Deployment

`docker build -f docker/Dockerfile -t vora .` fetches pinned models and builds the index; `docker run -p 8000:8000 vora` serves `http://localhost:8000`. Other devices need HTTPS (`scripts/make_cert.sh`); set `VORA_ACCESS_KEY` when the port is reachable from outside. It needs a 64-bit x86 or arm64 CPU, about 1.2 GB of disk for models, and 465 MB of memory for ASR, RAG and TTS plus 653 MB for the LLM. Main dependencies: sherpa-onnx, onnxruntime, llama-cpp-python, FAISS, FastAPI.

Proof	Result
arm64 image (native)	built, ready in 84 s, WebSocket smoke test passes in English, Chinese and with a real voice
amd64 image (emulated)	built 1.06 GB, boots, smoke test passes
Pi 4 shape (`--cpus=4 --memory=1g`, fast M2 cores)	first audio p50 979 / p95 2811 ms; Linux USS 465 MB
AWS t4g.small (2 Graviton2 vCPU, 2 GB)	HTTPS on 443, access key required; 3.8 to 5.6 s per answer: works, does not meet G1
Pi-class CPU (a1.xlarge) and Jetson	not run: AWS free plan refused the instance type, and there is no Jetson hardware

8. Limitations

- G5 faithfulness is 89%, below the 95% target. The misses are retrieval near-misses and the 0.5B model paraphrasing wrongly; off-topic refusal on the blind set is 60%. On real phone speech it is far lower, because the ASR text is wrong before retrieval starts.
- Raspberry Pi and Jetson are unmeasured. The closest evidence is the 4-core Docker run on M2 cores, which is optimistic.
- The Chinese voice has an unknown licence and a MOS proxy of 3.44, below the target. An Apache-2.0 swap exists (aishell3, 40 MB) but is over the 30 MB TTS limit.
- Real microphones, Firefox and physical phones were not tested automatically. The VORA Box question sets are self-authored, unlike the real-voice results.
- Cantonese is not supported.